

1. For each matrix equality, give the elementary coefficient matrix that enables to go from the right to the left. You can either multiply by the elementary matrix on the left or on the right but not on both sides.

(a) ($C_1 \leftrightarrow C_3$)

$$\begin{pmatrix} 1 & 2 & 4 & 3 & 9 \\ 2 & -4 & -1 & 1 & -5 \\ -1 & 1 & 3 & -6 & 3 \\ 1 & 2 & 1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix} \begin{pmatrix} 4 & 2 & 1 & 3 & 9 \\ -1 & -4 & 2 & 1 & -5 \\ 3 & 1 & -1 & -6 & 3 \\ 1 & 2 & 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix}$$

(b) ($L_2 \leftrightarrow L_3$)

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ -1 & -6 & 3 & 1 & 3 \\ 2 & 1 & -1 & -4 & -5 \\ 1 & 3 & 4 & 2 & 9 \end{pmatrix} = \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ 2 & 1 & -1 & -4 & -5 \\ -1 & -6 & 3 & 1 & 3 \\ 1 & 3 & 4 & 2 & 9 \end{pmatrix} \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix}$$

(c) ($C_4 \leftarrow 5C_4$)

$$\begin{pmatrix} 1 & 1 & 1 & 10 & 2 \\ 2 & 1 & -1 & -20 & -5 \\ -1 & -6 & 3 & 5 & 3 \\ 1 & 3 & 4 & 10 & 9 \end{pmatrix} = \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix} \begin{pmatrix} -3 & 1 & 1 & 2 & 2 \\ 12 & 1 & -1 & -4 & -5 \\ -5 & -6 & 3 & 1 & 3 \\ -10 & 3 & 4 & 2 & 9 \end{pmatrix} \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix}$$

(d) ($C_1 \leftarrow C_1 - C_4$)

$$\begin{pmatrix} -1 & 1 & 1 & 2 & 2 \\ 8 & 1 & -1 & -4 & -5 \\ -4 & -6 & 3 & 1 & 3 \\ -8 & 3 & 4 & 2 & 9 \end{pmatrix} = \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ 2 & 1 & -1 & -4 & -5 \\ -1 & -6 & 3 & 1 & 3 \\ 1 & 3 & 4 & 2 & 9 \end{pmatrix} \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix}$$

(e) ($L_4 \leftarrow L_4 + 2L_2$)

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ 2 & 1 & -1 & -4 & -5 \\ -1 & -6 & 3 & 1 & 3 \\ 5 & 5 & 2 & -6 & -1 \end{pmatrix} = \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ 2 & 1 & -1 & -4 & -5 \\ -1 & -6 & 3 & 1 & 3 \\ 1 & 3 & 4 & 2 & 9 \end{pmatrix} \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix}$$

2. Give the inverse of the following matrices. If you think the matrix is not invertible, that might be possible.

(a)

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

(b)

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

(c)

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

(d)

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -4 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

(e)

$$\begin{pmatrix} 1 & 0 & 2 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

(f)

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

3. Number of operations for this algorithm. (Only first order term, some explanations needed.)

```
def cholesky(A):
    n = A.shape[0]
    for j in range(0,n):
        for i in range(j,n):
            for k in range(0,j):
                A[i,j] = A[i,j] - A[i,k] * A[j,k]
    A[j,j] = np.sqrt(A[j,j])
    for i in range(j+1,n):
        A[i,j] = A[i,j] / A[j,j]
```

Note:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \quad \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$$

Number of flops (floating-point operations) is

$$\left(\sum_{j=0}^{n-1} \sum_{i=j}^{n-1} \sum_{k=0}^{j-1} 2\right) + \left(\sum_{j=0}^{n-1} 1\right) + \left(\sum_{j=0}^{n-1} \sum_{i=j+1}^{n-1} 1\right)$$